A. METINGS EN EENHEDE – *MEASUREMENTS AND UNITS*

1. Skakel om na (a) meter (b) mg en (c) sekondes **sonder** om u sakrekenaar te gebruik:

Convert to (a) metre, (b) mg, (c) seconds ***without*** *using your calculator:*

(a) 795,3 cm = $795{,}3 \times 10^{-2}$ m $= 7{,}953 \times 10^{0}$ m 1 (1)

(b) 47,21 kg = $47{,}21$ mg $= 4{,}721 \times 10^{3}$ mg 1 (1)

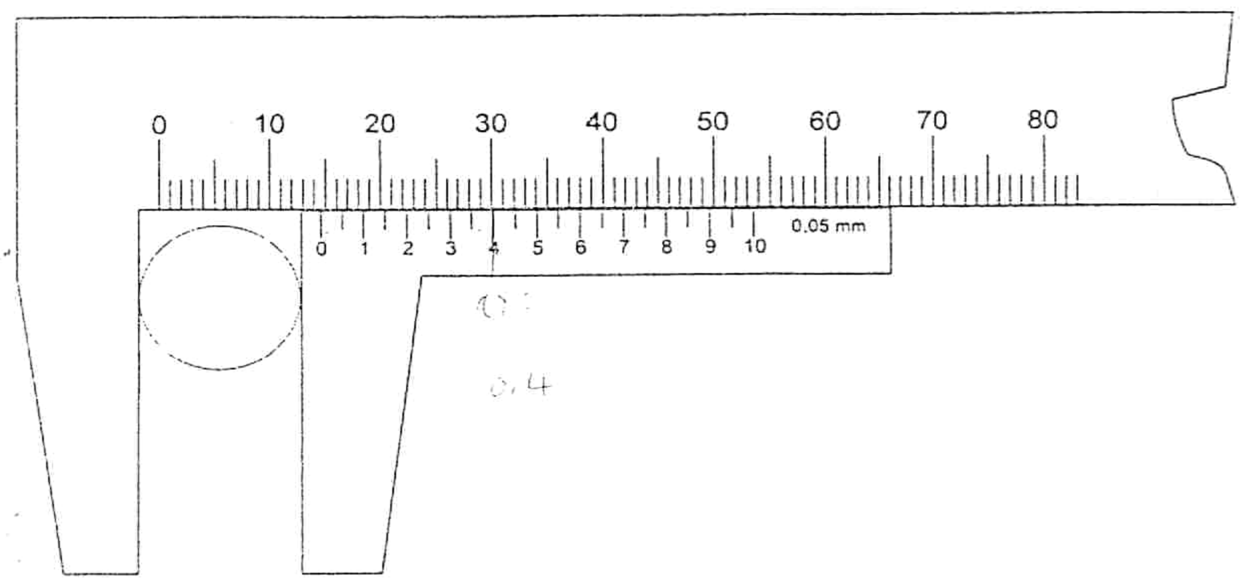
(c) 573,12 μs = $573{,}12 \times 10^{-6}$ s $= 5{,}7312 \times 10^{-4}$ s 1 (1)

2. Druk 1N uit in terme van kg, m en s. F = ma

Write 1N *in terms of* kg, m *and* s. kg m/s^{-2} 1 (1)

3. 'n Student meet die deursnee van 'n stafie met 'n skuifpasser. Die skets toon die lesing op die skuifpasser. Aanvaar dat die skuifpasser geen nulpuntfout het nie.

A student uses vernier callipers to measure the diameter of a rod. The sketch shows the reading on the vernier callipers. Assume that the vernier callipers have no zero error.



3.1 Wat is die deursnit van die stafie volgens die skuifpasserlesing in mm?

What is the diameter of the rod according to the vernier calliper in mm?

14,4 mm 1 (1)

3.2 Wat is die kleinste indeling op die skuifpasser in die skets in mm?

What is the smallest reading on the vernier calliper in the sketch in mm?

0,05 mm 1 (1)

[6]

B1 STATIESE EWEWIG – *STATIC EQUILLIBRIUM*

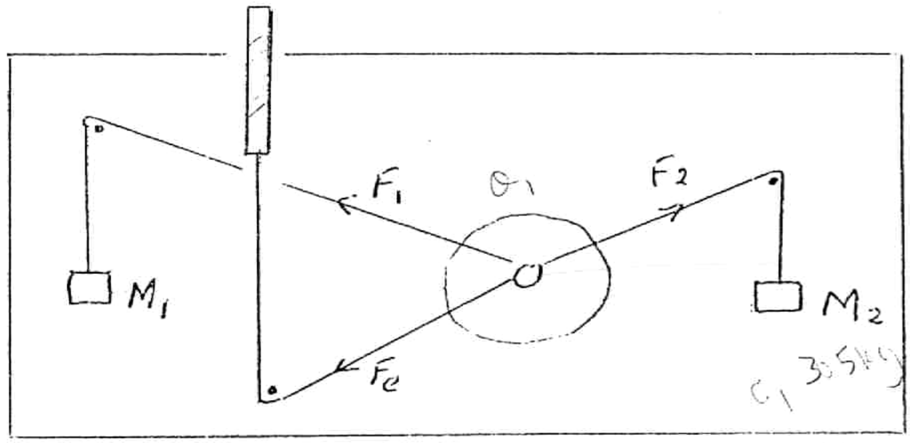


Fig B1

Kragtediagram*:*

Diagram of forces:

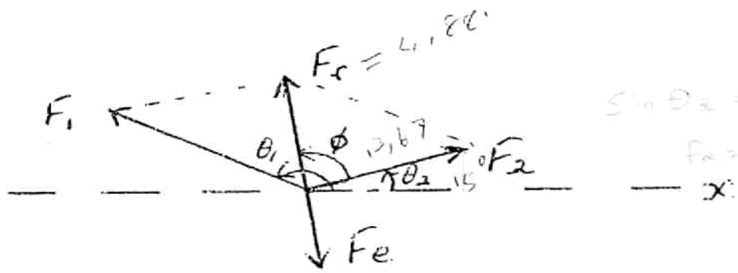


Fig B2

Die volgende is bekend:

The following are known:

$M_2 = 305$ g

$\theta_2 = 15°$

$\phi = 13,67°$

$|\bar{F}_r| = 4,88N$

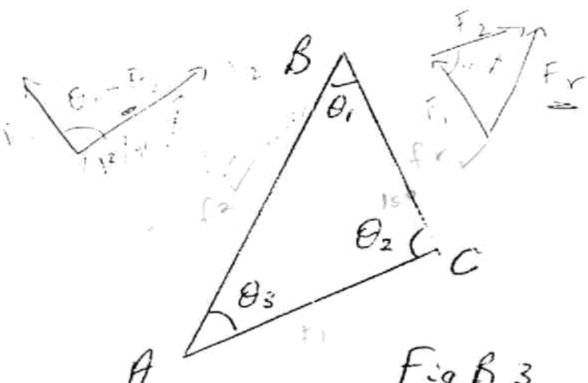


Fig B3

Sinusreël:

Sice law:

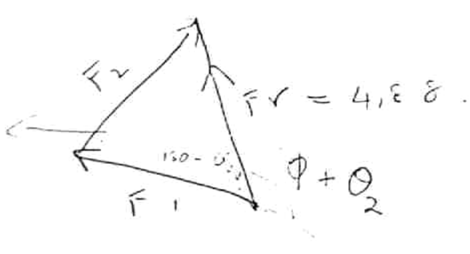
$$\frac{\sin\theta_1}{AC} = \frac{\sin\theta_2}{AB} = \frac{\sin\theta_3}{BC}$$

Cosinus reel:

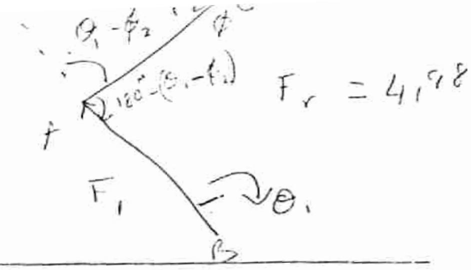
Cosine law: $AC^2 = BC^2 + AB^2 - 2(BC)(AB)\cos\theta_1$

F_r is die resultant vn F_1 en F_2

F_r *is the resultant of* F_1 *and* F_2



$-\theta_2$



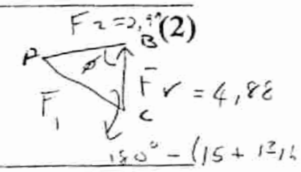
1. Bewys dat die grootte van krag $F_2 = 2,99$ N

Prove that force $F_2 = 2,99$ N

$$\frac{\sin\theta_1}{F_2} = \frac{\sin 163,33}{F_r}$$

$$F_2 = \frac{F_r \sin\theta_1}{\sin 163,33} =$$ 0

(2)



2. Bereken die grootte van krag F_1 in N.

Calculate the magnitude of force F_1 *in* N.

$\cancel{B^2 = A^2 + C^2 +}$

$$AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos\phi$$

$$F_1 = \sqrt{(2,99)^2 + (4,88)^2 - 2(2,99)(4,88)\cos 13,6}$$

$$= 2,097 \text{ N}$$ 4

(4)

3. Bereken die rigting van F_1 met die x-as, met ander woorde θ_1 (Fig. B_2).

Determine the direction of F_1 *relative to the* x-*axis, that means determine* θ_1 *(Fig.* B_2*)*

(4)

Maak gebruik van die volgende driehoek en parallelogram van kragte.

Utilize the following triangle and parallelogram of forces.

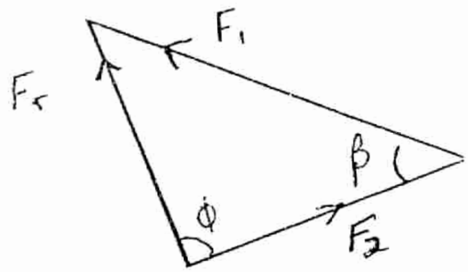


Fig B4

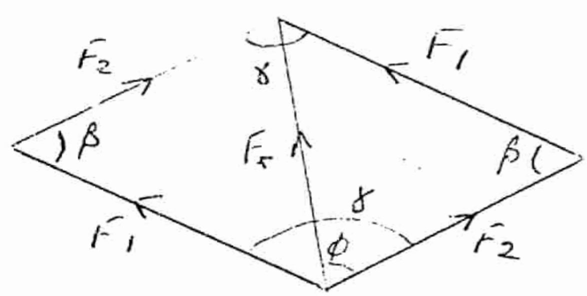


Fig B5

$\theta_2 = 15°$

6

(4)

[10]

B2 STATIESE EWEWIG – *STATIC EQUILLIBRIUM*

'n Horisontale staaf, lengte ℓ, is in ewewig met twee eksterne kragte wat daarop inwerk, naamlik

F_1 = 3,0 N opwaarts aan die linkereindpunt teen 'n hoek van 20° met die horisontaal, en
F_2 = 2,0 N afwaarts op die regtereindpunt teen 'n hoek van -23° met die horisontaal.

Die staaf word gestut (steunpunt) $\frac{1}{3}\ell$ vanaf die linkereindpunt.

A horizontal rod, length ℓ is in equilibrium with two external forces, namely:
F_1 = 3,0 N *upwards on the left-side end making an angle* 20° *with the horizontal, and*
F_2 = 2,0 N *downwards on the right-side end, making n angle* -23° *with the horizontal..*

The rod is suspended (at the pivot point) $\frac{1}{3}\ell$ *from the left-side end.*

1. Gee drie toestande vir ewewig van die staaf. / *Give three conditions for the equilibrium state of the rod.*

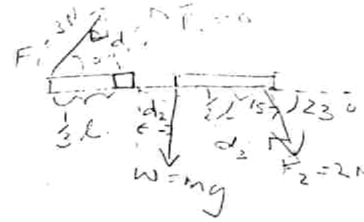
The rod must be in rotational equilibrium : That is the sum of the torques = 0 $[\Sigma \tau = 0]$
The rod must be in conditional equilibrium That is the sum of all forces acting on the object should be = 0 $\therefore$ $\Sigma F_i = 0$

(3)

$\tau = F \times d$

$\sin 20^\circ = \frac{d_1}{(\frac{1}{3})\ell}$ $\quad \frac{1}{2}\ell$

$d_1 = \frac{1}{3}\ell \sin 20^\circ$



2. Bereken die massa van die staaf. / *Calculate the mass of the rod.* ($g = 9{,}8\ ms^{-2}$).

$\bar{R} = 0 \quad \therefore -3 \times \frac{1}{3}\ell \sin 20^\circ - m(9{,}8)(\frac{1}{6}\ell) - 2 \times \frac{5}{6}\ell \sin 157^\circ = 0$

$\ell - \frac{1}{3}\ell$

$\frac{\ell - 2\ell}{6}$

$m\left(\frac{9{,}8}{6}\ell\right) = -\ell\sin 20^\circ - \frac{5}{3}\ell \sin 157^\circ$

$\cdot \ell$

$m = 0{,}608\ kg$

$= \sin 157^\circ$

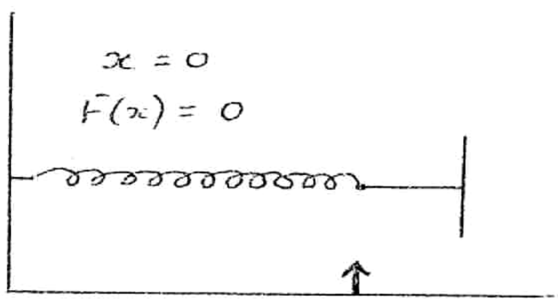
(7)

[10]

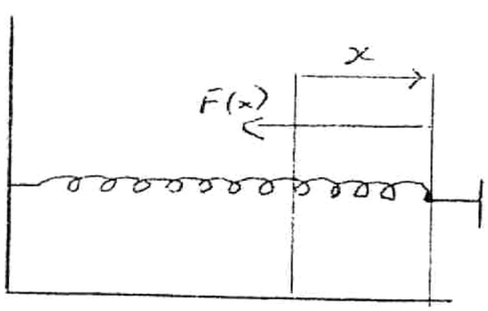
C1 ELASTISITEIT - *ELASTICITY*

$\frac{5}{6}\ell \sin 157^\circ$

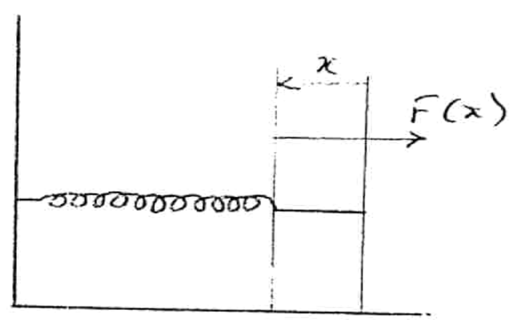
$\frac{1}{2}\ell + \frac{1}{3}\ell = \frac{3\ell + 2\ell}{6} = \frac{5}{6}\ell$



(a)



(b)



(c)

In Fig (a) is 'n veer met 'n aangehegte handvatsel in 'n ontspanne toestand, in (b) is die veer gerek oor 'n verlenging x en in (c) is die veer saamgedruk met 'n afstand x. Let op die herstelkrag F(x) in alle gevalle.

In Fig (a) *a spring with an attached handle is illustrated in its relaxed state. In Fig* (b) *the spring is stretched by* a *an increased length* x *and in(c) the spring is compressed by* x. *Take note of the restoring force* F(x) *in all cases.*

1. As 'n krag van 4,9 N aangewend word om die veer te rek vanaf sy ontspanne posisie, rek dit 12 mm.

$F = -kx$

If a force of 4,9 N *is applied to stretch the spring the increase in length is* 12mm.

(a) Wat is die veerkonstante? / *What is the spring constant?*

$F = 4,9$ N $\quad x = 12$mm $= 0,012$ m

$F = +kx$

$k = +\frac{F}{x} = \frac{4,9}{0,012} = +408,3$ N/m

4

(4)

(b) Watter grootte krag sal op jou hand uitgeoefen word as jy die veer 17 mm uitrek?

What will be the magnitude of the force exerted on your hand if you stretch the spring by 17 mm?

$x = 17$mm $= 0,017$ m

$F = -kx$

$= -(408,3)(0,017)$

$= -6,9411$ N

4

(4)
[8]

C2 ELASTISITEIT - ***ELASTICITY***

Die onderstaande table toon data verkry uit 'n eksperiment op 'n volkome elastiese veer.

The data in the table below represent experimental results of an elastic spring.

M (kg)	x (m)	F (N)
0	0	0
1	0,117	9,81
2	0,234	19,62
3	0,350	29,43

(4)

Verwerk die data (bepaal F) en bepaal die massa van 'n voorwerp wat die veer 300 mm sal laat uitrek.

Process the data (determine F) *and calculate the mass of an object which will extend the spring by* 300 mm.

($g = 9{,}81\ ms^{-2}$)

		Extension	Force (N) = ma
When M = 1:	1 (9,81)	= 0,117	9,81
When M = 2:	2 (9,81)	= 0,234	19,62
When M = 3:	3 (9,81)	= 0,350	29,43
When extension = 0,3 m:	m	= 0,3	

$F = -kx$ $k = \frac{F}{x}$

$m = \frac{3(9{,}81) \times 0{,}3}{0{,}35} = 25{,}2$ kg

(0)

(10)

D EENVOUDIGE HARMONIESE BEWEGINGS - *SIMPLE HARMONIC MOTIONS*

'n Vyftiggram massa is vas aan 'n veer. Die massa is laat los op 'n punt 30 mm vanaf sy ewewigsposisie. Die ossillasies wat volg is waargeneem by die punt van ewewig.

A 50 g hanger was suspended from the end of a spring. The hanger was released at a displacement of 30 mm from its equilibrium position. The oscillations were observed around the equilibrium position.

1. Wat is die amplitude van die ossillasies? / *What is the amplitude of the oscillations?*

The amplitude of the oscillation is the maximum displacement that ~~make~~ takes place between oscillation, from the equilibrium position In this case Amplitude = 30 mm = 0,03 m

(2) (2)

2. Sal die amplitude oor 'n lang tyd konstant bly? Motiveer.

Will the amplitude remain constant over a long period? Motivate.

No, because the amplitude of the oscillation becomes smaller due to the air resistance. Therefore as time continues, amplitude decreases. ~~in~~

(2) (2)

3. By watter posisie sal die spoed

At which position will the speed be

(a) Maksimum / *maximum*

(b) Zero

(a) The speed will be maximum at the top

(b) The speed will be zero at the middle. 0

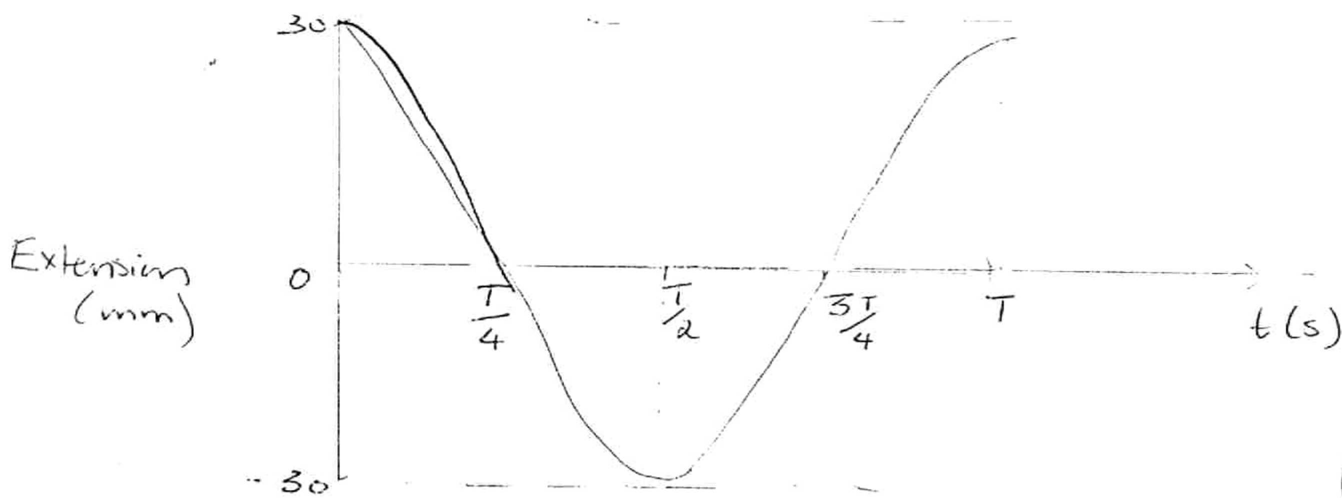
(2)

4. Maak 'n vryhandskets wat die verplasing y van die massa vanaf sy punt van ewewig toon as 'n funksie van tyd

$t = \frac{T}{4}$, $\frac{T}{2}$, $\frac{3T}{4}$ en T. Aanvaar dat by t = 0 die verplasing gelyk was aan die amplitude (30 mm).

Make a free hand sketch which shows the displacement y *of the mass from its equilibrium position as a function of time* $t = \frac{T}{4}$, $\frac{T}{2}$, $\frac{3T}{4}$ and T. *Assume that at the instant* $t = 0$ *the displacement was equal to the amplitude* (30 mm).

Graph of Extension vs t



4

(4)

[10]

E. EKSPONENSIËLE VERVAL - *EXPONENTIAL DECAY*

Die resultate in die onderstaande table verteenwoordig "gooie" (t) en aantalle blokkies wat **nie** met die gekleurde kant na bo beland het nie (N_w)

The results in the table below represent numbers of throws (t) and the number of cubes which did not land with the painted side upwards N_w.

Throws / gooie (t)	N_w	ℓn Nw
0	100	4,61
1	82	4,41
2	66	4,19
3	56	4,03
4	48	3,87
5	40	3,69
6	34	3,53
7	27	3,295
8	21	3,04
9	17	2,83
10	12	2,48
11	10	2,30
12	10	2,30
13	10	2,30
14	8	2,07
15	7	1,95

2

1. Voltooi die tabel.
   Complete the table. **(2)**

2. Trek 'n grafiek van N_w teen t. (Kyk vir grafiekbladsy op bladsy 12)
   Draw a graph of N_w *versus* t. (*See graphic paper on page* 12). **(4)**

3. Definieer halveertyd ($T_{½}$) van 'n funksie.
   Define half-life of a function.
   The half life of a function is the period of time after which only half the number of ~~orig~~ original elements eg nuclie or dice remain.
   2
   (2)

4. Bepaal die halveertyd uit die grafiek.
   Determine the half life from the graph.
   half life will be where Nw = 4,61 ÷ 2 = 2,305
   $t_{\frac{1}{2}}$ = 13
   0
   (2)

5. Definieer die vervalkonstante (γ) van 'n funksie.

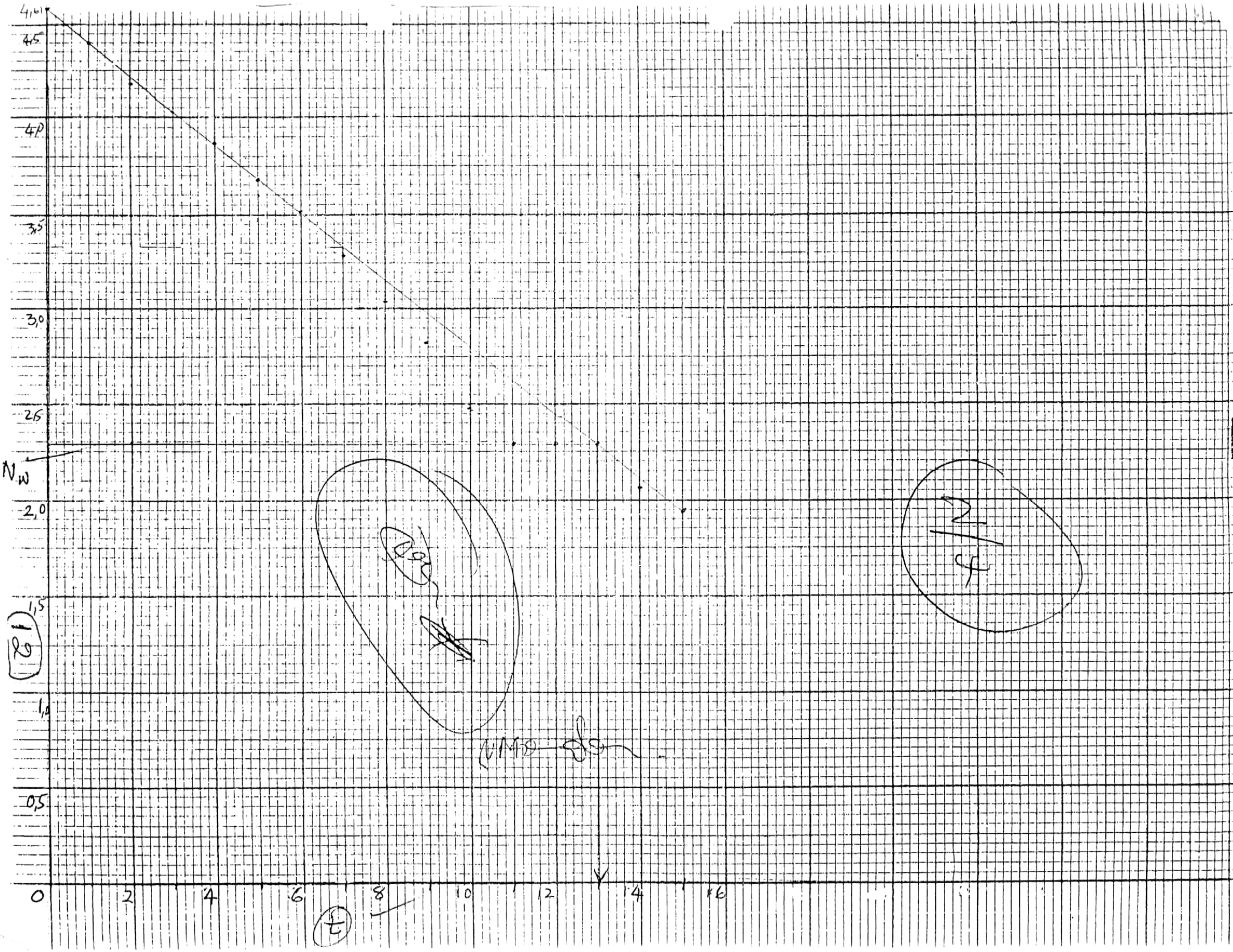
Define the decay constant (γ) of a function.

The decay constant, is a constant value defined by $\frac{N_{\omega}}{t}$, that ~~chan~~ changes the value of the function, with change in time.

0

(2)

[12]


4,61
4,5
4,0
3,5
3,0
2,5
N_W
2,0
1,5
1,0
0,5
0
2
4
6
8
10
12
14
16
t
2/7
12